Research Paper List

OpenCV-Based System for Crack Identification and Width Estimation

Basic to Advanced • Paper titles • Publication details • DOI / links • Topic relevance

Purpose

A curated bibliography and topic-wise reading guide for a project involving crack identification, image preprocessing, segmentation, crack area/length/width estimation, calibration, evaluation, and deep-learning comparisons.

Important note

This PDF lists paper titles and bibliographic links only; it does not reproduce or create the research papers. Some entries are search-topic placeholders rather than verified individual publications. Check the publisher record for exact author names, issue/pages, and publication year before citing any paper in academic work.

Reading sequence

1. Reviews and fundamentals → 2. Traditional image processing → 3. Crack geometry and width measurement → 4. Datasets and deep learning → 5. Advanced methods and targeted literature searches.

1. Fundamentals and Literature Reviews

No.	Paper title / publication	Year / venue	DOI / link	Relevance
1	**Crack Detection Using Image Processing: A Critical Review and Analysis** Alexandria Engineering Journal	2018	https://doi.org/10.1016/j.aej.2017.01.020	Traditional image processing, segmentation, feature extraction, and limitations.
2	**Review on Computer Vision-Based Crack Detection and Quantification Methodologies for Civil Structures** Construction and Building Materials	2022	https://doi.org/10.1016/j.conbuildmat.2022.129238	Crack detection, segmentation, quantification, and research gaps.
3	**Computer Vision Framework for Crack Detection of Civil Infrastructure—A Review** Engineering Applications of Artificial Intelligence	2023	https://doi.org/10.1016/j.engappai.2022.105478	Computer-vision pipeline, datasets, algorithms, and evaluation.
4	**A Review of Computer Vision-Based Crack Detection Methods in Civil Infrastructure: Progress and Challenges** Remote Sensing	2024 Remote Sensing	https://doi.org/10.3390/rs16162910	Traditional and deep-learning methods, challenges, and trends.
5	**Image Processing Techniques for Concrete Crack Detection: A Scientometrics Literature Review** Remote Sensing	2023 Remote Sensing	https://doi.org/10.3390/rs15092400	Research trends and citation mapping.
6	**A Focused Review of Recent Advances in Deep Learning and Computer Vision for Concrete Crack Detection and Measurement** Discover Civil Engineering	2026 Discover Civil Engineering	https://doi.org/10.1007/s44290-026-00618-4	Recent detection and measurement research.
7	**Advanced Image Processing and Deep Learning Methods for Automated Crack Detection in Civil Engineering Structures: A Comprehensive Review** Journal/review record—verify publisher metadata	2026	https://doi.org/10.1007/s42107-026-01758-9	Image-processing and deep-learning developments.

2. Traditional Image Processing and Segmentation

No.	Paper title / publication	Year / venue	DOI / link	Relevance
8	**A Robust Automatic Crack Detection Method from Noisy Concrete Surfaces** Machine Vision and Applications	2010	https://doi.org/10.1007/s00138-009-0244-5	Filtering, thresholding, and handling noisy concrete surfaces.
9	**CrackTree: Automatic Crack Detection from Pavement Images** Pattern Recognition Letters	2012 Pattern Recognition Letters	https://doi.org/10.1016/j.patrec.2011.11.004	Pavement crack detection and complex crack representation.
10	**Image-Based Crack Detection for Real Concrete Surfaces** Publication details should be checked	2008	https://doi.org/10.1002/tee.20244	Traditional image analysis on concrete surfaces.
11	**Accurate Extraction and Measurement of Fine Cracks from Concrete Block Surface Image** Search by title to confirm full publication metadata	2002	https://scholar.google.com/scholar?q=%22Accurate+extraction+and+measurement+of+fine+cracks%22	Fine crack extraction and early measurement methods.

3. Crack Width, Length, Area, and Geometric Measurement

No.	Paper title / publication	Year / venue	DOI / link	Relevance
12	**Practical Image Measurement of Crack Width for Real Concrete Structure** Electronics and Communications in Japan	2009	https://doi.org/10.1002/ecj.10151	Physical crack-width measurement from images.
13	**Determination of Geometric Parameters of Cracks in Concrete by Image Processing** Advances in Civil Engineering	2019 Advances in Civil Engineering	https://doi.org/10.1155/2019/2398124	Geometric features and crack measurements.
14	**Image-Based Automated Width Measurement of Surface Cracking** Sensors, 21(22), 7534	2021 Sensors, 21(22), 7534	https://doi.org/10.3390/s21227534	Automated width measurement and image geometry.
15	**Image Analysis Method for Crack Distribution and Width Estimation for Reinforced Concrete Structures** Automation in Construction	2018 Automation in Construction	https://doi.org/10.1016/j.autcon.2018.03.012	Crack distribution and width estimation.
16	**Practical Application of Digital Image Processing in Measuring Concrete Crack Widths in Field Studies** Practice Periodical on Structural Design and Construction	2023	https://doi.org/10.1061/SC.1943-5576.0000735	Field measurements and practical validation.
17	**Concrete Crack Width Measurement Using a Laser Beam and Image Processing Algorithms** Applied Sciences, 13(8), 4981	2023 Applied Sciences, 13(8), 4981	https://doi.org/10.3390/app13084981	Physical references and measurement calibration.
18	**Neighborhood Shortest Distance Method for Concrete Crack Width Detection in Images** Engineering Structures	2025 Engineering Structures	https://doi.org/10.1016/j.engstruct.2024.119519	Complex crack geometry and width estimation.
19	**Segment-Based Adaptive Neighborhood Orthogonal Projection Method for Automatic Complex Concrete Crack Width Measurement** Results in Engineering	2025 Results in Engineering	https://doi.org/10.1016/j.rineng.2025.108020	Advanced width estimation for branching cracks.

4. Datasets and Deep Learning

No.	Paper title / publication	Year / venue	DOI / link	Relevance
20	**SDNET2018: An Annotated Image Dataset for Non-Contact Concrete Crack Detection Using Deep Convolutional Neural Networks** Data in Brief, 21, 1664–1668	2018 Data in Brief, 21, 1664–1668	https://doi.org/10.1016/j.dib.2018.11.015	Dataset design and crack/no-crack benchmarking.
21	**DeepCrack: Learning Hierarchical Convolutional Features for Crack Detection** IEEE Transactions on Image Processing	2019	https://doi.org/10.1109/TIP.2018.2878966	Hierarchical CNN features and pixel-level crack detection.
22	**DeepCrack: A Deep Hierarchical Feature Learning Architecture for Crack Segmentation** Neurocomputing	2019 Neurocomputing	https://doi.org/10.1016/j.neucom.2019.01.036	Multi-scale features and crack segmentation.

5. Advanced Methods and Recent Research

No.	Paper title / publication	Year / venue	DOI / link	Relevance
23	**A Review of Computer Vision-Based Crack Detection Methods in Civil Infrastructure: Progress and Challenges** Remote Sensing	2024 Remote Sensing	https://doi.org/10.3390/rs16162910	Recent methods and multimodal approaches.
24	**Neighborhood Shortest Distance Method for Concrete Crack Width Detection in Images** Engineering Structures	2025 Engineering Structures	https://doi.org/10.1016/j.engstruct.2024.119519	Advanced geometric width measurement.
25	**A Focused Review of Recent Advances in Deep Learning and Computer Vision for Concrete Crack Detection and Measurement** Discover Civil Engineering	2026 Discover Civil Engineering	https://doi.org/10.1007/s44290-026-00618-4	Recent methods and measurement challenges.
26	**Advanced Image Processing and Deep Learning Methods for Automated Crack Detection in Civil Engineering Structures: A Comprehensive Review** Journal/review record—verify publisher metadata	2026	https://doi.org/10.1007/s42107-026-01758-9	Current image-processing and deep-learning research.

6. Additional Search Topics for Expanding the Literature Review

No.	Paper title / publication	Year / venue	DOI / link	Relevance
27	**Crack detection using CLAHE and morphological BlackHat enhancement** Search topic—identify individual papers	Various years	https://scholar.google.com/scholar?q=concrete+crack+detection+CLAHE+BlackHat+morphological+image+processing	Directly relevant to contrast enhancement and dark-crack enhancement.
28	**Otsu thresholding and adaptive thresholding for concrete crack segmentation** Search topic—identify individual papers	Various years	https://scholar.google.com/scholar?q=concrete+crack+segmentation+Otsu+adaptive+thresholding	Thresholding approaches for binary crack masks.
29	**Skeletonization and distance transform for crack width estimation** Search topic—identify individual papers	Various years	https://scholar.google.com/scholar?q=crack+width+estimation+skeletonization+distance+transform	Centreline extraction and local width estimation.
30	**Pixel-to-millimetre calibration for image-based crack measurement** Search topic—identify individual papers	Various years	https://scholar.google.com/scholar?q=image+based+concrete+crack+width+measurement+calibration+pixels+millimetres	Physical calibration and measurement reliability.
31	**Crack segmentation using U-Net and semantic segmentation** Search topic—identify individual papers	Various years	https://scholar.google.com/scholar?q=concrete+crack+segmentation+U-Net+semantic+segmentation	Advanced pixel-level segmentation.
32	**Concrete crack detection using vision transformers** Search topic—identify individual papers	Various years	https://scholar.google.com/scholar?q=concrete+crack+detection+vision+transformer	Transformer-based approaches and current methods.

Recommended keywords for further literature search

concrete crack detection image processing • CLAHE BlackHat crack enhancement • Otsu thresholding crack segmentation • morphological image processing concrete cracks • crack contour area estimation • skeletonization crack length estimation • distance transform crack width estimation • pixel-to-millimetre calibration crack measurement • crack width measurement validation • SDNET2018 concrete crack dataset • DeepCrack crack segmentation • U-Net concrete crack segmentation • vision transformer crack detection • crack severity classification structural health monitoring

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